Submodule 33.05: Handling Outliers (Visual Detection & Statistical Bounds) (Hinglish)

Topic Overview & Motivation

Outliers woh unusual data points hote hain jo normal customer distribution se bahut door hote hain. Unsupervised learning mein outliers kaafi harmful hote hain:

- **K-Means Centroid Distortion**: K-Means centroid calculate karne ke liye mean use karta hai. Ek single extreme outlier centroid ko apni taraf kheench leta hai aur cluster structure bigad deta hai.
- **PCA Distortion**: PCA maximum variance ki direction dhundhta hai. Agar ek outlier $666,666 par baitha ho, toh PCA component usi outlier ki direction align ho jayega aur baki sabhi 2,239 points compress ho jayenge.

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1. Pairplot se Outlier Detection

Outliers ko visually dekhne ke liye Ma'am ne numerical columns ka pairplot banaya: `cols = ["Income", "Recency", "Response", "Age", "Total_Spending", "Total_Children"]`

Reference: smartcart.ipynb (Cell 22).

```
cols = ["Income", "Recency", "Response", "Age", "Total_Spending", "Total_Children"
sns.pairplot(df_cleaned[cols])
plt.show()
```

Pairplot se Mili Key Insights:

1. **Age Outliers**: Kuch customers ki age 120 saal se upar dikh rahi thi (Birth year 1893, 1899 etc.). Yeh clearly invalid data entry errors hain.
2. **Income Outliers**: Ek customer ki income \$666,666 thi jo normal population (\$20k se $160k) se bilkul alag akele scatter plot mein stand out kar rahi thi.

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2. Statistical Filtering Bounds

Ma'am ne do logical bounds define kiye:

1. $\text{Age} < 90$: Data entry errors wale unrealistic ages ko remove kiya.
2. $\text{Income} < 600,000$: $666,666 wale isolated financial outlier ko remove kiya.

Crucial Engineering Tip:

Har step par dataframe ka state preserve karna (`df_cleaned` , `df_encoded` , `X_scaled`) best practice hai. Agar in-place replace karoge aur baad mein debug karna pada, toh pura notebook dobara chalana padega.

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3. Python Code Implementation

Reference: smartcart.ipynb (Cell 23).

```
print("Data size with outliers:", len(df_cleaned)) # 2240

# Outliers ko remove karna
df_cleaned = df_cleaned[(df_cleaned["Age"] < 90)]
df_cleaned = df_cleaned[(df_cleaned["Income"] < 600_000)]

print("Data size without outliers:", len(df_cleaned)) # 2236
```

Result:

- Exactly **4 outliers** drop hue (2,240 $\rightarrow$ 2,236 rows).
- Ab dataset completely clean aur outlier-free hai.